SC4001 Neural Networks and Deep Learning

Part A: HDB Resale Price Regression

Hu Han — Matriculation Number: U2320036H

Generative AI Declaration

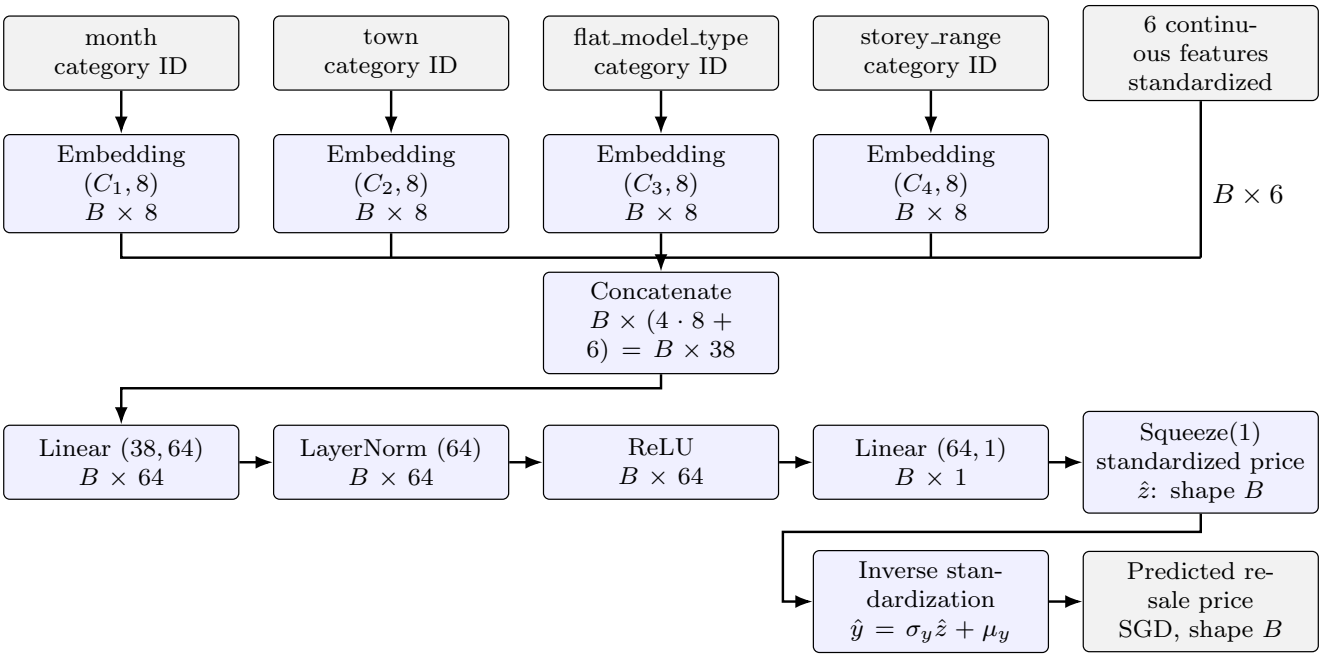
I used **Codex** to assist with the generation of some of the code segments during the development of this assignment. The tool was used primarily to generate initial code templates and to obtain suggestions for implementing specific programming tasks. All AI-generated code was carefully reviewed, modified where necessary, integrated into the overall solution by me, and tested to ensure it met the assignment requirements. I verified the correctness of the generated code through manual inspection, compilation/execution testing, debugging, and comparison with the assignment specifications and relevant course materials. The final submitted work represents my own understanding and effort. I take full responsibility for the accuracy, functionality, and academic integrity of the submitted solution.

Generative AI Tool Used: Codex
Purpose of Use: Programming assistance, README.md generation, and LaTeX formatting
Verification Method: Manual code review, testing, debugging, and validation against assignment requirements

A1

(a) Block Diagram & Parameter Count

Block Diagram:



Explanation:

- Each categorical embedding has dimension 8.
- The hidden layer has width 64.
- B denotes the batch size.
- C_j counts the known training categories for feature j, plus one extra category with ID 0 for unknown values.

Number of trainable parameters:

Each categorical feature j uses an embedding table with C_j rows and 8 trainable values per row. We add one extra category for unknown category for each feature, which gives:

$$C_j = \text{number of categories in feature } j + 1.$$

Calculate C_j for every categorical features:

Categorical feature	Number of training categories	C_j calculation
`month`	12	$C_1 = 12 + 1 = 13$
`town`	26	$C_2 = 26 + 1 = 27$
`flat_model_type`	43	$C_3 = 43 + 1 = 44$
`storey_range`	17	$C_4 = 17 + 1 = 18$

Six continuous features: `dist_to_nearest_stn`, `dist_to_dhoby`, `degree_centrality`, `eigenvector_centrality`, `remaining_lease_years`, and `floor_area_sqm`.

Each contributes one input value. Concatenating the four 8-dimensional embeddings with these six values gives the total input dimension D:

$$D = 4 \times 8 + 6 = 38.$$

Layer-by-layer trainable parameter counts:

Component	Calculation	Trainable parameters
Four embedding tables	$(13 + 27 + 44 + 18) \times 8$	816
Linear $(38, 64)$	$38 \times 64 + 64$	2,496
LayerNorm (64)	$64 + 64$	128
Linear $(64, 1)$	$64 \times 1 + 1$	65
Total	$816 + 2{,}496 + 128 + 65$	**3,505**

Explanation:

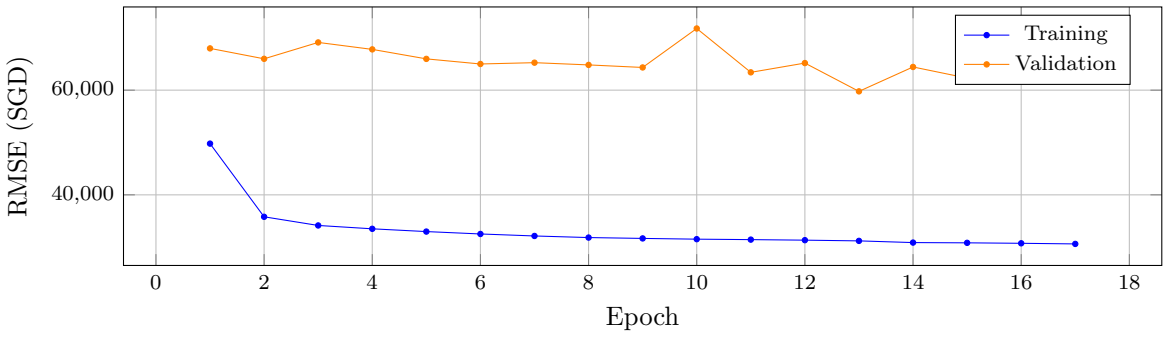
- Each linear layer includes a weight matrix and a bias vector.
- LayerNorm has 64 learnable scale parameters and 64 learnable shift parameters.
- Concatenation, ReLU, squeezing, and inverse target standardization add no trainable parameters.

(b) Hyperparameter Search

- **Search method:** Ray Tune grid search over 25 combinations: Hidden-layer width $h \in \{16, 32, 64, 128, 256\}$ and embedding dimension $d \in \{2, 4, 8, 12, 16\}$ shared by all four categorical features.
- **Data split:** Training on 2017–2020 and validation on 2021.
- **Training settings:** Seed 42; Adam with learning rate 0.01; batch size 512; maximum 20 epochs; early-stopping patience 4 and minimum validation RMSE improvement 10^{-4}.
- **Best result:** $h = 128$, $d = 12$; validation RMSE 59,766.91 SGD at epoch 13. Training stopped at epoch 17 and epoch-13 weights were retained.

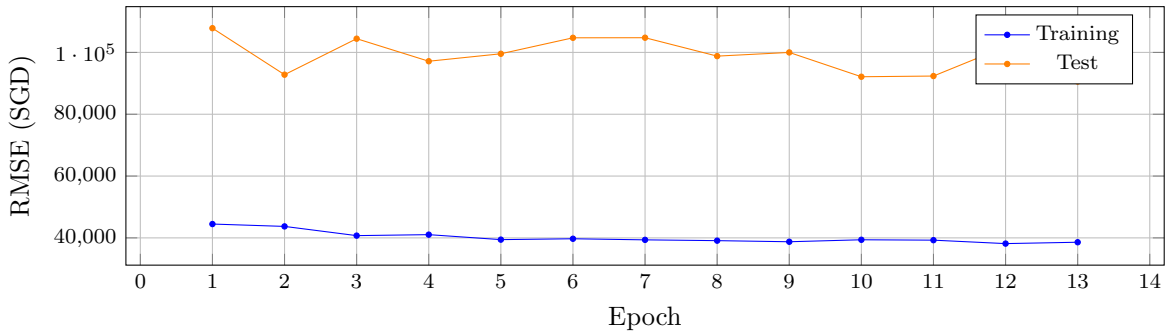
(c) Selected Configuration

- **Optimal hyperparameters:** Hidden-layer width 128; embedding dimension 12 for each categorical feature.
- **Best validation RMSE:** 59,766.91 SGD at epoch 13.
- **Learning curves:** Training and validation RMSE are shown below.



(d) Retraining and Test Performance

- **Retraining:** A fresh model with the selected architecture was trained on 2017–2021 (116,427 records), with preprocessing refitted on this combined set. Seed 42 and the A1(b) optimiser settings were retained.
- **Epoch count:** Fixed at 13 using the best validation epoch from A1(c).
- **Evaluation:** Training and testing RMSE were computed after each epoch in SGD. The 2022 test set contained 26,702 records.
- **Final test performance:** RMSE = 90,595.88 SGD; $R^2 = 0.716868$.



(e) Discussion

- **Configuration choice:** $(h, d) = (128, 12)$ achieved the lowest validation RMSE (59,766.91 SGD), improving on the original $(64, 8)$ configuration (63,422.46 SGD) by 5.76%. The runner-up $(256, 2)$ scored 59,926.48 SGD, only 159.58 SGD higher.
- **Capacity:** Larger widths and embeddings did not consistently improve validation RMSE. For example, at $h = 128$, increasing d from 12 to 16 worsened the best RMSE to 65,599.97 SGD.
- **Overfitting evidence:** After epoch 13, training RMSE declined from 31,207.16 to 30,627.12 SGD while validation RMSE rose from 59,766.91 to 69,319.56 SGD by epoch 17, supporting early stopping and indicating overfitting in later epochs.
- **Generalisation:** Retraining ended with training RMSE 38,591.78 SGD and test RMSE 90,595.88 SGD. This gap indicates limited generalisation; changes between the training years and 2022 may also contribute.

A2

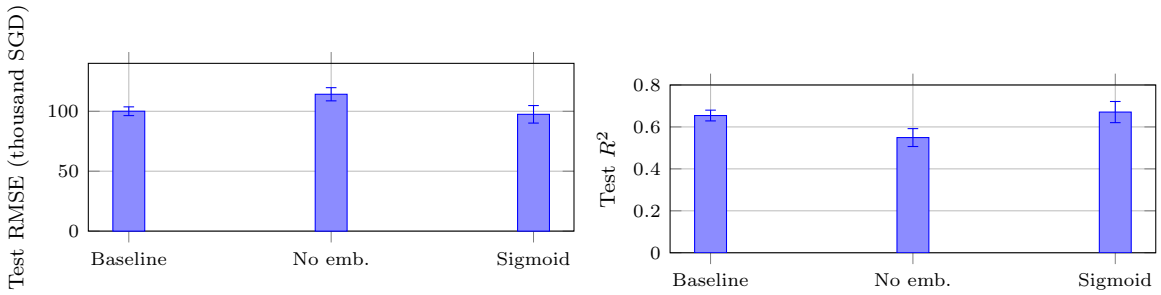
- **Protocol:** All three models were trained from scratch on 2017–2021 for 13 epochs with $h = 128$, $d = 12$ where applicable, Adam (learning rate 0.01), and batch size 512. Preprocessing was fitted on the combined training set.
- **Seeds and evaluation:** Seeds $\{42, 43, 44, 45, 46\}$ were used for each model, with identical minibatch orders for matching seeds. Each run was evaluated once on the 2022 test set. Results are mean $\pm$ sample SD $(n = 5)$; error bars show SD. The independent shuffle generator differs from A1(d), so its seed-42 result can differ.

Per-seed results: RMSE values are in SGD; all runs used 13 epochs.

Model	Seed	Training RMSE	Test RMSE	Test R^2
Baseline	42	37,926.73	98,124.06	0.6679
Baseline	43	38,258.40	97,431.21	0.6725
Baseline	44	37,931.61	101,068.42	0.6476
Baseline	45	37,476.46	97,601.34	0.6714
Baseline	46	38,618.59	106,030.81	0.6122
Without embeddings	42	54,406.94	114,211.19	0.5500
Without embeddings	43	55,308.87	117,697.22	0.5221
Without embeddings	44	55,730.65	119,940.38	0.5037
Without embeddings	45	53,933.65	113,674.84	0.5542
Without embeddings	46	54,487.32	105,507.78	0.6160
Sigmoid	42	38,004.36	96,090.96	0.6815
Sigmoid	43	39,493.07	109,142.47	0.5891
Sigmoid	44	38,728.61	89,514.55	0.7236
Sigmoid	45	36,920.54	98,480.10	0.6654
Sigmoid	46	38,093.72	94,102.55	0.6945

Five-seed summary:

Model	Test RMSE (SGD)	Test R^2
Baseline	$100,051.17 \pm 3,652.75$	0.6543 ± 0.0256
Without embeddings	$114,206.28 \pm 5,500.29$	0.5492 ± 0.0427
Sigmoid	$97,466.13 \pm 7,310.91$	0.6708 ± 0.0504



(a) Removing Categorical Embeddings

- **Modification:** All four embedding tables were removed. The first layer changed from $54 \to 128$ to $6 \to 128$; LayerNorm, ReLU, and the output layer were retained. Trainable parameters decreased from 8,649 to 1,281.
- **Result:** Mean test RMSE increased by 14.15%, and mean R^2 decreased from 0.6543 to 0.5492. RMSE worsened for four of five matched seeds.
- **Interpretation:** Categorical inputs provide useful information beyond the continuous features. Mean training RMSE also rose from 38,042.36 to 54,773.49 SGD, consistent with reduced fitting ability. This ablation removes both information and parameters, so their effects cannot be separated.

(b) Replacing ReLU with Sigmoid

- **Modification:** Only the hidden activation changed to Sigmoid.
- **Result:** Mean test RMSE decreased by 2.58%, and mean R^2 increased to 0.6708. However, RMSE SD increased from 3,652.75 to 7,310.91 SGD; Sigmoid improved RMSE for three of five matched seeds.
- **Interpretation:** Sigmoid changes the bounded hidden representation and gradient behaviour while leaving capacity unchanged in parameter count. Mean training RMSE was similar (38,248.06 versus 38,042.36 SGD). Its modest average test improvement and larger seed variation do not establish a reliable advantage over ReLU in this experiment.

A3

(a) Selecting λ

- **Architecture:** Both branches share the 54-dimensional input. The deep branch retains the A1 MLP ($h = 128$, $d = 12$); the wide branch is a single $54 \to 1$ linear layer. Outputs are combined as $\mathrm{MLP}(x) + \lambda\,\mathrm{Linear}(x)$ before inverse target scaling.
- **Search protocol:** $\lambda \in \{0, 0.25, 0.5, 1, 2\}$; seed 42; training on 2017–2020 and validation on 2021. Adam, learning rate 0.01, batch size 512, maximum 20 epochs, and early-stopping patience 4 with minimum improvement 10^{-4} were retained. All candidates used identical initial weights and minibatch orders.
- **Selection criterion:** Lowest validation RMSE achieved during training. Results for all five candidates are shown below.

λ	Best validation RMSE (SGD)	Best epoch	Epochs run
0	59,720.34	4	8
0.25	60,935.69	4	8
0.5	62,760.50	4	8
1	62,025.90	6	10
2	63,258.46	6	10

- **Optimal value:** $\lambda = 0$, with validation RMSE 59,720.34 SGD at epoch 4; training stopped at epoch 8. The linear branch contributes nothing at this value, so the selected predictor reduces to the baseline. Positive coefficients did not improve validation performance in this search.

(b) Five-Seed Evaluation

- **Protocol:** The selected $\lambda = 0$ model was retrained from scratch on 2017–2021 using seeds $\{42, 43, 44, 45, 46\}$. All runs retained the A1 settings and fixed 13-epoch duration, with preprocessing fitted on the combined set. Each final model was evaluated once on the 2022 test set.
- **Baseline comparison:** The five A1-architecture baseline runs recorded in A2 were reused with matching seeds, preprocessing, optimiser settings, epochs, and minibatch orders. The original A1(d) single-seed result is also shown as a separate reference.

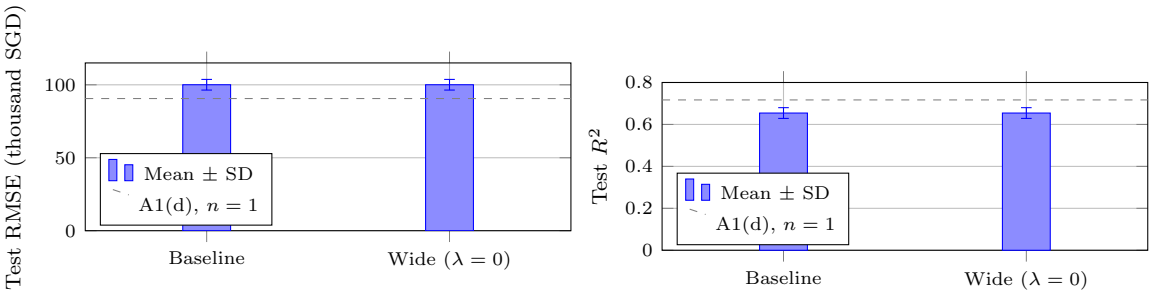
Per-seed results: RMSE values are in SGD.

Seed	Training RMSE	Test RMSE	Test R^2
42	37,926.73	98,124.06	0.6679
43	38,258.40	97,431.21	0.6725
44	37,931.61	101,068.42	0.6476
45	37,476.46	97,601.34	0.6714
46	38,618.59	106,030.81	0.6122

Comparison: Five-seed results are mean ± sample SD ($n = 5$).

Model	Test RMSE (SGD)	Test R^2
A1(d) baseline ($n = 1$)	90,595.88	0.7169
Baseline ($n = 5$, A2)	$100,051.17 \pm 3,652.75$	0.6543 ± 0.0256
Wide-and-deep ($\lambda = 0$)	$100,051.17 \pm 3,652.75$	0.6543 ± 0.0256

Plots: Bars show five-seed means with sample SD; dashed lines show the A1(d) single-seed reference.



- **Result:** Mean test RMSE was 100,051.17 ± 3,652.75 SGD and mean R^2 was 0.6543 ± 0.0256. All five paired results and the trained embedding/MLP weights matched the baseline exactly, as expected when $\lambda = 0$ disables the wide contribution. The selected model therefore provides no improvement over the matched baseline.
- **A1 reference:** A1(d) reported 90,595.88 SGD and $R^2 = 0.7169$ for one seed with a different shuffle protocol. Its lower RMSE cannot be attributed to the architecture difference; the matched five-seed comparison isolates that difference.

(c) Discussion

Mechanism: The linear branch provides a shorter gradient path to the shared embeddings, which may improve optimisation when gradients through the MLP are attenuated. Due to its simplicity, it can learn additive patterns alongside the MLP's nonlinear relationships, potentially improving generalisation.

Evidence: However, all positive λ values produced higher best validation RMSE than $\lambda = 0$, meaning these results provide no evidence of a benefit from the linear branch under the tested settings.

Limitations: Gradient norms were not measured, so potential relief of vanishing gradients cannot be verified.

A4

(a) FeatureAblation Computation

- **Models:** The A1-architecture baseline and selected wide-and-deep model ($\lambda = 0$) used seed 42, identical preprocessing, and 13 epochs on 2017–2021. The matched baseline checkpoint from A2 and wide checkpoint from A3(b) were loaded without retraining.
- **Feature groups:** Captum FeatureAblation was applied to all 26,702 test samples. Each categorical ID was ablated as one original feature, replacing its entire embedding; each of the six continuous features was ablated separately. Predictions and attributions were in SGD.
- **References:** Categorical features used combined-training modes: month 7, town SENGKANG, flat model "4 ROOM, Model A", and storey range "04 TO 06". Continuous features used zero after standardisation, corresponding to their combined-training means. Both models used the same references.
- **Aggregation:** For model m, sample i, and feature j, attribution and global importance were computed as

$$a_{ij}^{(m)} = f_m(x_i) - f_m\left(x_i^{(j \leftarrow r_j)}\right), \qquad I_j^{(m)} = \frac{1}{26{,}702} \sum_{i=1}^{26{,}702} \left| a_{ij}^{(m)} \right|,$$

  where r_j is the reference value. Signed per-sample attributions were saved for both models.

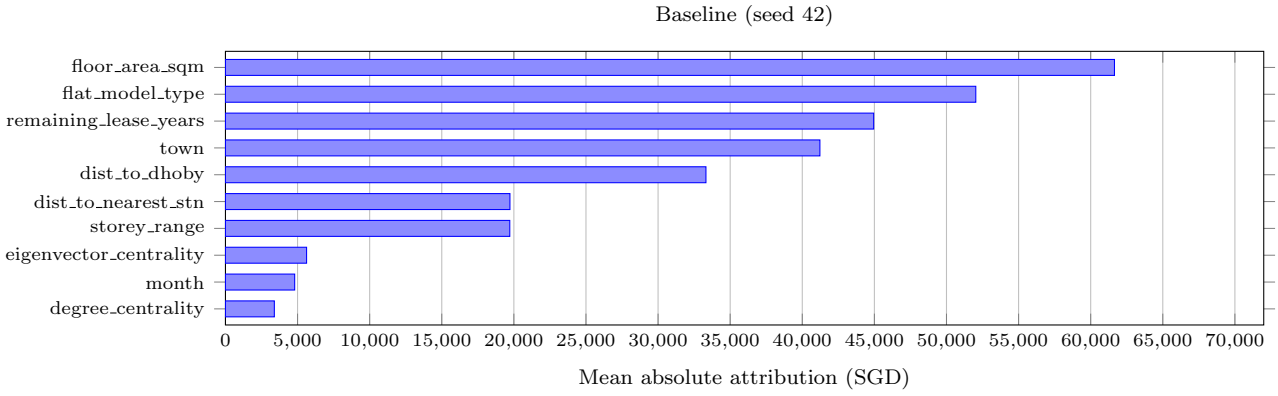
Global importance scores: Mean absolute attribution (SGD).

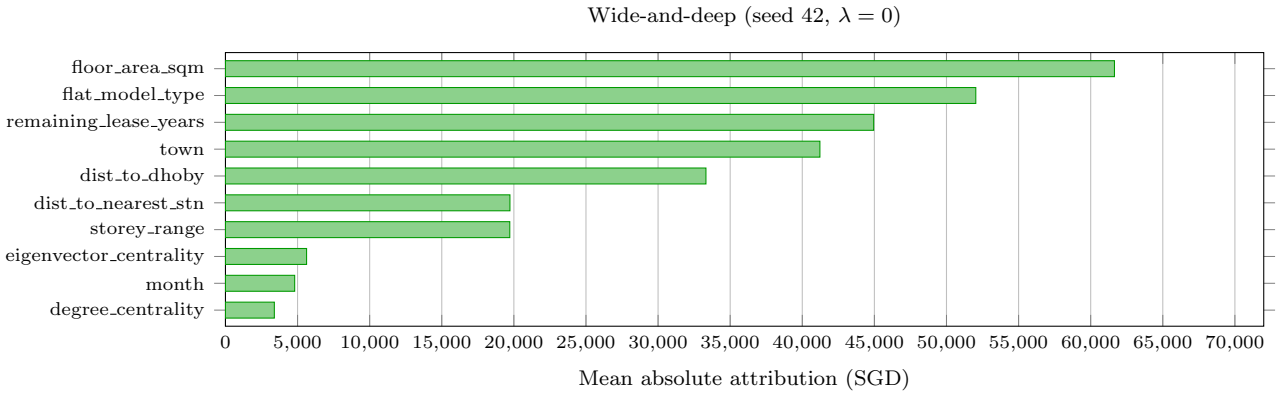
Original feature	Baseline	Wide-and-deep
month	4,800.15	4,800.15
town	41,222.37	41,222.37
flat_model_type	52,032.36	52,032.36
storey_range	19,716.84	19,716.84
dist_to_nearest_stn	19,727.62	19,727.62
dist_to_dhoby	33,319.83	33,319.83
degree_centrality	3,389.72	3,389.72
eigenvector_centrality	5,625.70	5,625.70
remaining_lease_years	44,949.20	44,949.20
floor_area_sqm	61,648.57	61,648.57

Verification: All ten Captum attributions were checked against manual prediction differences on eight samples. Both full attribution matrices were finite and had shape 26,702 × 10; their maximum absolute difference was 0 SGD, consistent with $\lambda = 0$ and matching trained weights.

(b) Feature Importance Plots

Ordering: Each plot shows all ten features in descending mean absolute attribution (SGD). The two rankings and scores coincide because the selected $\lambda = 0$ model is equivalent to the matched baseline.





(c) Interpretation

- **Top three features:** `floor_area_sqm` (61,648.57 SGD), `flat_model_type` (52,032.36 SGD), `remaining_lease_years` (44,949.20 SGD).
- **Comparison and discussion:** The importance scores show that these three features are the most important in predicting the price of a property.
- **Limitation of the analysis:** Different reference values used for feature ablation may affect the relative importance of continuous features, changing the results.